



Case Study: Target Corporation Data Breach

LESSON 1

A real-world examination of one of the most significant retail data breaches in history — and what it teaches us about cybersecurity fundamentals.

Background

Putting everything covered so far into one real story. In **November 2013**, the busiest shopping period of the year in the United States, attackers installed malware on the point-of-sale systems in over **1,800 Target retail stores**. By mid-December the breach was discovered. By then the damage was done.

- ⚠ This breach occurred during the peak holiday shopping season — a deliberate choice by the attackers to maximise the volume of card transactions being processed.



The Numbers

The scale of the Target breach was staggering. These figures represent real people, real financial harm, and real organisational consequences.

40M

Card Records Stolen

Credit and debit card records exfiltrated from POS systems across
1,800+ stores.

70M

Customer Records Exposed

Names, addresses, phone numbers, and email addresses
compromised.

\$162M

Estimated Total Cost

Total financial impact to Target Corporation in USD.

18M

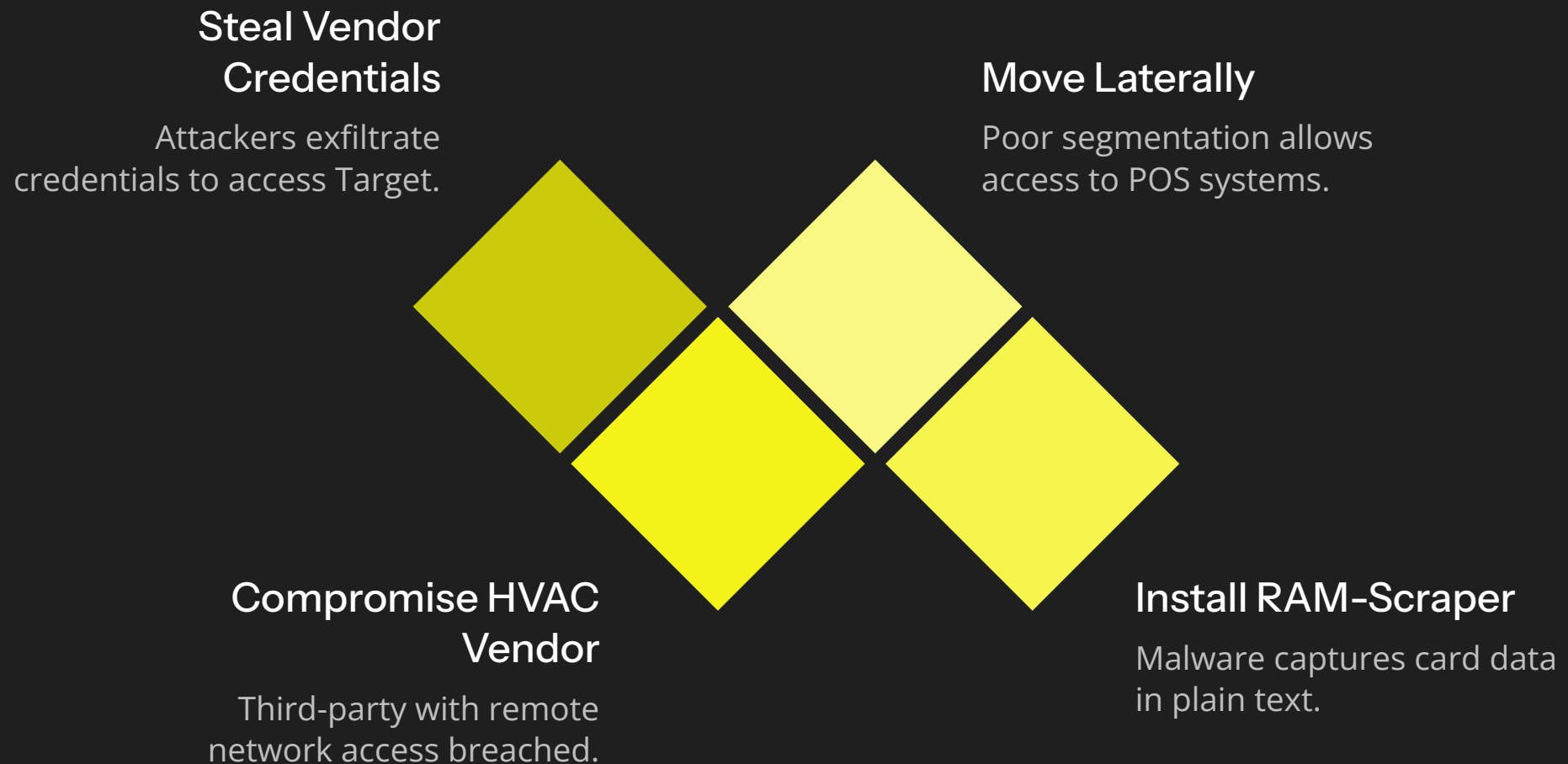
Cards Sold Criminally

Stolen cards sold on criminal marketplaces after the breach.

⊗ Target's CEO and CIO both resigned in the aftermath. The human cost of a security failure extends far beyond the technical.

How It Happened

The attackers did not breach Target directly. The attack unfolded in a carefully staged sequence, exploiting trust, access, and architectural weaknesses at every step.



Each stage of this attack exploited a different failure — a compromised third party, stolen credentials, absent network segmentation, and unencrypted data in memory. No single fix would have been sufficient; defence in depth was required at every layer.

CIA Triad Impact Analysis

Analysing this breach through the lens of the CIA triad reveals how all three pillars of information security were violated.

Confidentiality — Violated

40 million card numbers and PINs were exfiltrated and accessed by unauthorised persons. Data that should have remained private was exposed to criminal actors at scale.

Integrity — Violated

The RAM-scraping malware intercepted and modified data flows in transit, capturing card data without authorisation at the point of processing. Data was tampered with before it could be protected.

Availability — Impacted

Card issuers cancelled millions of cards after the breach, disrupting legitimate transactions for customers. Target's revenue dropped significantly in the quarter following public disclosure.



Root Cause & Lessons Learned

Four distinct failures compounded to create the conditions for this breach. Each maps directly to a control domain covered in this course.

1

Third-Party Vendor Risk

Insufficient vetting and monitoring of vendor access. The HVAC contractor was granted remote network access without adequate controls or oversight. *Who was accountable for that?*

2

Poor Network Segmentation

Vendor access should never have reached POS systems. The absence of segmentation between the vendor access network and the retail POS network was a critical technical control gap.

3

Alerting Failure

Target's security team received warnings from FireEye malware detection but did not act on them. Detective controls only work if you act on them — detection without response is worthless.

4

Incident Response Failure

The breach ran for **three weeks** before action was taken. This reflects weak SOC and CIRT integration — a failure of process, not just technology.

Food for Your Brain

CRITICAL THINKING

Three questions to keep thinking about as you progress through this course. There are no simple answers — that is the point.

1 Which category of control would have most likely prevented this?

Consider preventive, detective, and corrective controls. Which single intervention would have had the greatest impact on stopping this breach before it began?

2 Target had detective controls in place (FireEye). Why did they not work?

The technology flagged the threat. The alerts were generated. So what failed? Think about people, process, and culture — not just tools.

3 What compensating controls could have reduced impact?

Even if the breach could not be fully prevented, what controls might have limited the damage — reducing the number of records stolen or the duration of the attack?